UNIT 8
ROBOTICS

Learning Objectives

- Identify key emerging technologies and their applications.
- Define robotics and explain its significance.
- Describe the use of robotics in manufacturing, drones, and surgery.
- List benefits of robotics, including safety, efficiency, and consistency.
- Identify challenges of robotics, such as unemployment, skill loss, and high costs.

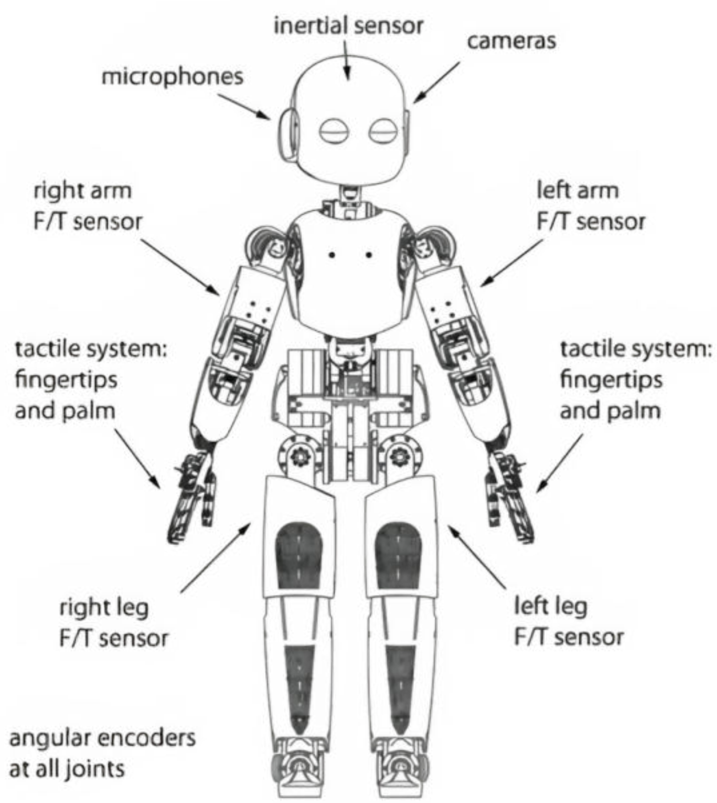
INTRODUCTION TO EMERGING TECHNOLOGIES

The world is advancing rapidly with the development of new technologies that are transforming how we live, work, and interact. Emerging technologies such as **Artificial Intelligence (AI), Biometrics, Vision Enhancement, Robotics, Quantum Cryptography, Computer-Assisted Translation (CAT), 3-D and Holographic Imaging, and Virtual Reality** are revolutionizing industries and everyday life.

These technologies have many applications, from improving productivity to creating safer and more environmentally friendly systems. However, as technology evolves at a remarkable pace, it's important to stay updated. Reviewing advancements on the internet every six months can help us keep up with these rapid changes.

WHAT IS ROBOTICS?

Robotics is one of the most exciting emerging technologies. Robots are machines designed to perform tasks automatically or with some level of human guidance. Robotics has been around for many years, primarily used in the manufacturing industry. Robots are capable of doing tasks that are repetitive, dangerous, or require a high level of precision.



Applications of Robotics

1. In Manufacturing:

Robots are widely used in manufacturing to improve efficiency and consistency. They perform tasks such as:

- Painting car bodies
- Welding parts of cars

- Manufacturing microchips and electrical goods
- Storing and retrieving items in automatic warehouses

Robots in manufacturing are controlled through:

- **Pre-programmed instructions:** The robot follows a sequence of commands to complete tasks, such as painting a car.
- **Manual guidance:** A human operator either guides the robot arm directly or uses sensors to perform tasks. The robot then saves these actions and repeats them automatically.

For example, robots equipped with sensors can ensure they don't spray paint when no car is present or stop working if the paint runs out. This makes them efficient and reliable in repetitive tasks.

2. Use of Drones

Drones are robotic devices that can fly without a pilot. They are used in:

·Military operations for reconnaissance and surveillance

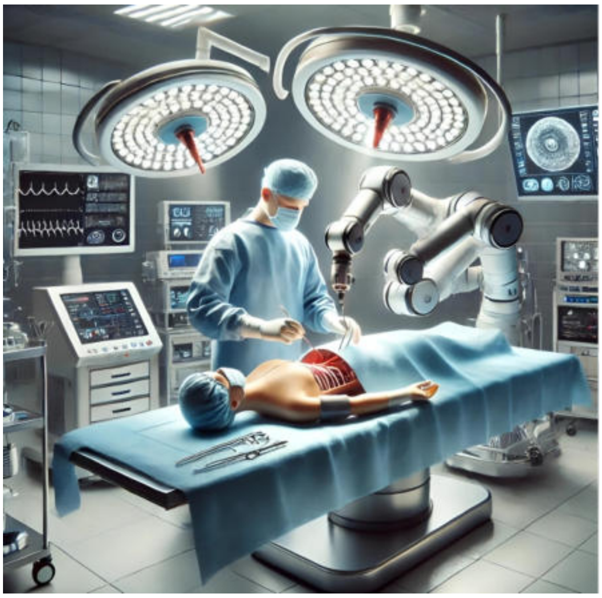
Civilian tasks such as mapping landscapes in 3-D, monitoring weather, and aiding in search and rescue missions during natural disasters.

Drones help in tasks that might be too dangerous or difficult for humans.

3. In Surgery

Robots are also used in the field of medicine, especially in surgeries. Robotic surgery enables:

- Precise and complex procedures
- Flexibility and control beyond standard techniques



Surgeons operate these robots, which are equipped with camera arms and mechanical joints that mimic human movement, making surgeries safer and more effective.

ADVANTAGES OF ROBOTICS

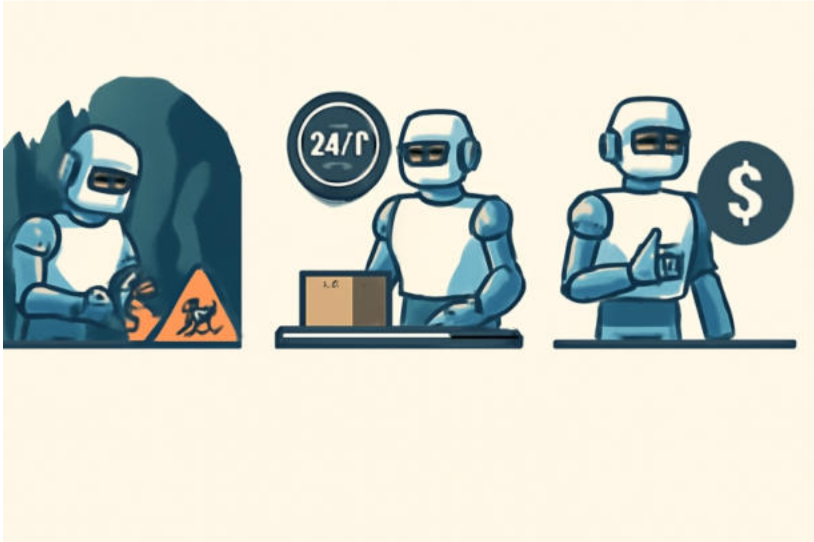
Robots offer several benefits, including:

Safety: They can work in dangerous environments harmful to humans.

Efficiency: Robots can work non-stop (24/7), unlike humans.

Cost-Effectiveness: Though expensive initially, robots are cheaper in the long term since they do not require wages or holidays.

Consistency: Robots produce uniform results, such as ensuring all cars on an assembly line are identical.
Time-Saving: They handle repetitive tasks, freeing humans for creative and skilled work like quality control or designing.

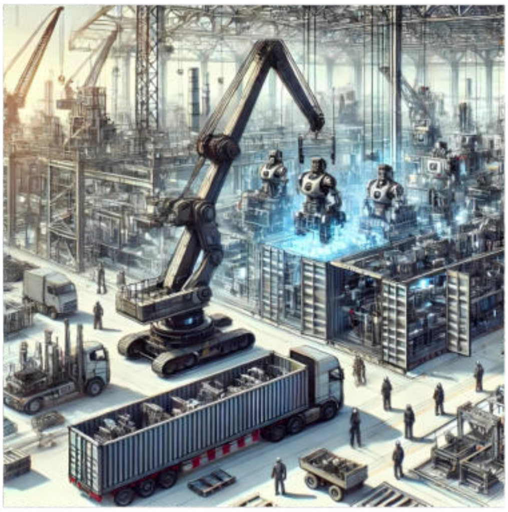


DISADVANTAGES OF ROBOTICS

While robots are beneficial, they also come with challenges:
Limited Flexibility: Robots struggle with unique or unusual tasks, such as creating custom glassware.
Unemployment: Replacing humans with robots can reduce jobs for skilled workers.

Skill Loss: Over-reliance on robots may lead to a decline in human skills like welding.

High Costs: Setting up and maintaining robots is expensive.
Job Relocation: Factories equipped with robots can be moved to other regions, impacting local employment.



EXERCISE 1

Choose the correct answer.

1. What is one common application of robotics in manufacturing?

A. Teaching
B. Painting car bodies
C. Cooking
D. Writing books

2. Which technology is used for flying without a pilot?

A. AI
B. Drones
C. Biometrics
D. Quantum cryptography

3. What is one advantage of robots?

A. They never need maintenance
B. They can work non-stop
C. They are very cheap to buy
D. They don't require programming

4. Where is robotic surgery commonly used?

A. Schools
B. Factories
C. Hospitals
D. Farms

5. What can drones be used for?

A. Cooking
B. Firefighting
C. Writing books
D. Welding

6. What is an example of a disadvantage of robots?

A They never get tired

B They are expensive to maintain

C They help in repetitive tasks

D They increase consistency

7. What does a robot use to understand its surroundings?

A Sensors

B Motors

C Cameras

D Wires

8. Which part of a robot mimics human movement in surgery?

A Legs

B Camera arm

C Interactive mechanical arms

D Wheels

9. Which technology helps in translation?

A Robotics

B CAT

C AI biometrics

D Virtual reality

10. What does ICT stand for?

A Information and Computer Technology

B Innovative Creative Tools

C Intelligent Circuit Technology

D Industrial Computing Techniques

EXERCISE 2

Fill in the blanks.

1. Robotics is an example of an ____________ technology.

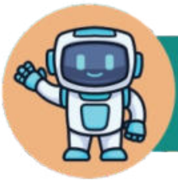
2. Drones are commonly used for ____________ missions in the military.

3. Robots can work ____________ hours without stopping.

4. Robotic surgery provides more ____________ than standard techniques.

5. A robot in manufacturing is often guided by ____________.

6.Robots are good at doing ____________ tasks.



EXERCISE 3

Mark "True" or "False".

	TRUE	FALSE
Robots never require programming.	☐	☐
Drones are used for search and rescue missions.	☐	☐
Robotics can help reduce human jobs in manufacturing.	☐	☐
Robots are unable to perform repetitive tasks.	☐	☐
Robots in surgery use camera arms to perform tasks.	☐	☐
Quantum cryptography is a type of robotics.	☐	☐

EXERCISE 4

Answer the following questions:

1. Define robotics in your own words.

2. Name two advantages of using robots in factories.

3. What is one use of drones in civilian tasks?

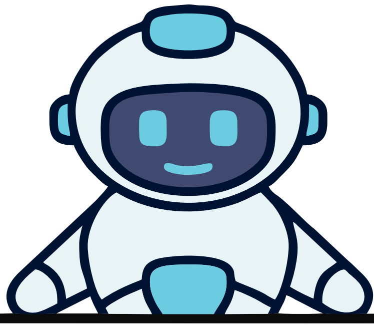
4. How do sensors help robots perform tasks?

5. Why might robots not be suitable for creating custom glassware?

6. Mention one disadvantage of robotics in manufacturing.

LAB ACTIVITY

WATCHING A VIDEO AND MAKING A ROBOT MODEL FROM CARDBOARD



Understand basic robot features by watching the video and create a robot model.

Activity Plan

1. Video Watching on "How Robots Work"

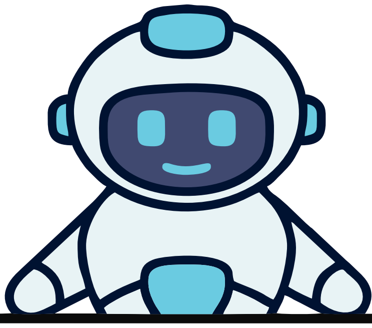
2. Hands-On Activity: Making a Robot Model from Carton

Discussion After Video:

- What tasks can robots perform better than humans?
- What are the main parts of a robot (e.g., sensors, motors, processing unit)?

LAB ACTIVITY

MAKING A ROBOT MODEL FROM CARDBOARD



Create a robot model using household materials.

Materials Needed:

- Cardboard boxes (small and medium-sized)
- Toilet paper or tissue rolls (for arms and legs)
- Plastic bottle caps (for wheels or buttons)
- Aluminum foil (for shiny robot parts)
- Glue, tape, scissors
- Markers or paint for decoration

Optional: LED lights or old wires (for decoration to make it look more "robotic")

Steps:

1.Prepare the Base:

- Use a medium-sized cardboard box as the robot's body.
- Attach a smaller box on top for the robot's head.

2.Add Features:

- Use tissue rolls or small cartons to create the robot's arms and legs. Attach them with glue or tape.
- Stick bottle caps on the head as "eyes" or "buttons."

3. Decorate:

- Wrap parts of the robot in aluminum foil to give it a metallic look.
- Draw or paint a control panel on the robot's body.

4. Optional Additions:

- Attach wires or small LED lights for a futuristic appearance.
- Create movable joints using split pins for a more dynamic model.

5. Show and Tell:

- Have each student display their robot and explain its name, design, and the tasks it can perform if it were real.